Exercise J-M

1. Let [x] denote the greatest integer $\le x$. where $x \in R$. If the domain of the real valued function $f(x) = \sqrt{\frac{|[x]|-2}{|[x]|-3}}$ is $(-\infty, a) \cup [b, c) \cup [4, \infty), a < b < c$, then the value of $a + b + c$ is : **[JEE(Main)July2021]**

माना [x] महत्तम पूर्णांक $\le$ x है, जहाँ $x \in R$ है। यदि वास्तविक मान फलन $f(x) = \sqrt{\frac{|[x]|-2}{|[x]|-3}}$ का प्रांत $(-\infty, a) \cup [b, c) \cup [4, \infty), a < b < c$, है, तो $a + b + c$ का मान है : **[JEE(Main)July2021]**

(A) 8 (B) 1 (C) –2 (D) –3

Ans. C

Sol. Put $|[x]| = t \quad t \in I \ , t \ge 0$

$$f = \sqrt{\frac{t-2}{t-3}}$$

$$\frac{t-2}{t-3} \ge 0$$

$$t \in [0,2] \cup (3,0)$$

$$\because t \in I$$

$$t \in [0, 2] \cup (4, \infty)$$

$[x] \in [-2, 2]$ or $[x] \in [4, \infty)$ or $[x] \in (-\infty, -4]$

$x \in [-2, 3)$ or $x \in [4, \infty)$ $x \in (-\infty, -3)$

$\Rightarrow x \in (-\infty, -3) \cup [-2, 3) \cup (4, \infty)$

$a = -3 \ \ b = -2, c = 3$

$a + b + c = -2$

2. Let $f : R - \left\{\frac{\alpha}{6}\right\} \to R$ be defined by $f(x) = \frac{5x+3}{6x-\alpha}$. Then the value of α for which (fof) (x) = x, for all $x \in R - \left\{\frac{\alpha}{6}\right\}$, is : **[JEE(Main)July2021]**

(A) 8 (B) 5 (C) 6 (D) No such α exists

माना $f : R - \left\{\frac{\alpha}{6}\right\} \to R$, $f(x) = \frac{5x+3}{6x-\alpha}$ द्वारा परिभाषित है। तो α का मान जिसके लिए, (fof) (x) = x, $\forall x \in R - \left\{\frac{\alpha}{6}\right\}$ है, है : **[JEE(Main)July2021]**

(A) 8 (B) 5 (C) 6 (D) ऐसे किसी α का अस्तित्व नहीं है

Ans. B

Sol. $f(f(x)) = x$

$$\frac{5f(x)+3}{6f(x)-2} = x$$

$5\,f(x) + 3 = 6x\,f(x) - x\,\alpha$

$$f(x) = \frac{-3-x\alpha}{5-6x} = \frac{x\alpha+3}{6x-5}$$

$$\frac{5x+3}{6x-\alpha} = \frac{x\alpha+3}{6x-5}$$

$30x^2 - 25x + 18x - 15 = 18x + 6\alpha\, x^2 - 3\alpha - x\alpha^2$

$6\alpha = 30$

$\alpha = 5$

3. Let [x] denote the greatest integer less than or equal to x. Then, values of $x \in R$ satisfying the equation $[e^x]^2 + [e^x + 1] - 3 = 0$ lie in the interval : **[JEE(Main)July2021]**

माना [x] महत्तम पूर्णांक $\le x$ है। तो समीकरण $[e^x]^2 + [e^x + 1] - 3 = 0$ को संतुष्ट करने वाली सभी वास्तविक संख्याऐं, x, निम्न में से किस अन्तराल में है ? **[JEE(Main)July2021]**

(A) $[1, e)$ (B) $[\log_e 2, \log_e 3]$ (C) $[0, 1/e)$ (D) $[0, \log_e 2)$

Ans. D

Sol. $[e^x]^2 + [e^x + 1] - 3 = 0$

$[e^x]^2 + [e^x] + 1 - 3 = 0$

Let $[e^x] = y$

$y^2 + y - 2 = 0$

$(y + 2)(y - 1) = 0$

$y = -2$ Not possible

$y = 1$

$[e^x] = 1$

$1 \le e^x < 2$

By applying log

$0 \le x < \ln 2$

$x \in [0, \ln 2)$

4. If [x] be the greatest integer less than or equal to x, then $\sum_{n=8}^{100}\left[\frac{(-1)^n n}{2}\right]$ is equal to :

[JEE(Main)July2021]

यदि [x] महतम पूर्णांक फलन दर्शाता है, तो $\sum_{n=8}^{100}\left[\frac{(-1)^n n}{2}\right]$ बराबर होगा : **[JEE(Main)July2021]**

(A) 0 (B) 4 (C) 2 (D) –2

Ans. B

Sol. $\sum_{n=8}^{100}\left[\frac{(-1)^n n}{2}\right]$

$4 - 5 + 5 - 6 + 6 \ldots\ldots\ldots\ldots 50 + 50 = 4$

5. Consider functions $f: A \to B$ and $g: B \to C\,(A, B, C \subseteq R)$ such that $(gof)^{-1}$ exists, then : **[JEE(Main)July2021]**

(A) f and g both are one-one (B) f is onto and g is one-one

(C) f and g both are onto (D) f is one-one and g is onto

फलनों $f: A \to B$ तथा $g: B \to C$ पर विचार कीजिए $(A, B, C \subseteq R)$ जिनके लिए $(gof)^{-1}$ अस्तित्व है, तो : **[JEE(Main)July2021]**

(A) f तथा g दोनों एकैकी है (B) f आच्छादक है तथा g एकैकी है

(C) f तथा g दोनों आच्छादक हैं (D) f एकैकी है तथा g आच्छादक है

Ans. D

Sol. $(gof)^{-1}$ exists $\Rightarrow$ gof is bijective

f(x) should be one-one

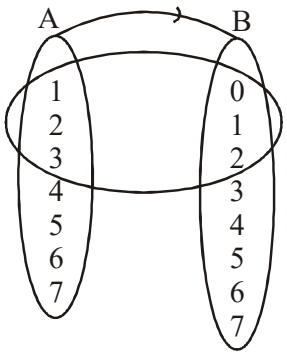
g(x) should be onto

6. Let $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then the number of bijective function $f: A \to A$ such that $f(1) + f(2) = 3 - f(3)$ is equal to ___. **[JEE(Main)July2021]**

माना : $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$ एक समुच्चय है। तो फलनों, $f: A \to A$, जो आच्छादक तथा एकैकी दोनों हैं तथा $f(1) + f(2) = 3 - f(3)$ को संतुष्ट करते हैं, संख्या बराबर है _____ **[JEE(Main)July2021]**

Ans. 720

Sol.



1, 2, 3 will be associated with 0, 1, 2 since f(1) + f(2) + f(3) = 3, then no. of method for these = $\lfloor 3$

Now 0, 4, 5, 6, 7 will be associated with 3, 4, 5, 6, 7 then no. of methods = $\lfloor 5$

total method = $\lfloor 5 \times \lfloor 3$

= 720

7. Let S = {1, 2, 3,4, 5, 6, 7}. Then number of possible functions $f: S \to S$ such that f(m.n) = f(m).f(n) for every m, n∈S and m.n ∈ S is equal to_____. **[JEE(Main)July2021]**

माना S = {1, 2, 3, 4, 5, 6, 7} है। तो ऐसे फलनों $f: S \to S$, जिनके लिए f(m.n) = f(m). f(n) ∀ m, n∈S तथा m.n ∈ S हैं, की संख्या बराबर है _____ **[JEE(Main)July2021]**

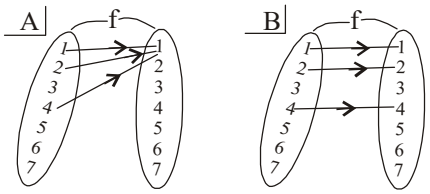
Ans. 490

Sol. f(m.n) = f(m). f(n) for ∀ m.n ∈ S

m = 1 : f(x) = f(1). f(x) ⇒ f(x) = 1

m = n = 2 : $f(4) = f^2(2) \Rightarrow$ f(2) = 1 & f(4) = 1

f(2) = 2 & f(4) = 4



m = 2, n = 3: f(6) = f(2). f(3)

for A | f(2) = 1 ⇒ f(6) = f(3) ∈ {1,2,3,4,5,6,7}

f(5) = f(7) ∈ {1,2,3,4,5,6,7} (can take any value)

Total = 7 × 7 × 7 = 343

for B | f(2) = 2 ⇒ f(6) = 2 f(3)

f(3)	f(6)
1	2
2	4
3	6

Total = 3 × 7 × 7 = 147

Total functions = 343 + 147 = 490

8. Let f : N → N be a function such that f(m + n) = f(m) + f(n) for every m, n ∈ N. If f(6) = 18, then f(2) . f(3) is equal to : **[JEE(Main)Aug.2021]**

माना f : N → N एक फलन है, जिसके लिए f(m + n) = f(m) + f(n) m, n ∈ N है। यदि f(6) = 18 है, तो f(2) . f(3) बराबर है : **[JEE(Main)Aug.2021]**

(A) 36 (B) 54 (C) 6 (D) 18

Ans. B

Sol. f(m + n) = f(m) + f(n)

m = n = 1

$f(2) = 2f(1)$

$f(3) = f(2) + f(1)$

$f(3) = 3f(1)$

$f(2)\, f(3) = 6\,(f(1))^2$

$m = 3 \quad n = 3$

$f(6) = 2f(3)$

$18 = 2f(3)$

$f(3) = 9$

$3f(1) = 9 \Rightarrow f(1) = 3$

$f(2)\, f(3) = 6 \times 9 = 54$

9. The range of the function $f(x) = \log_{\sqrt{5}}\left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right)\right)$ is :

[JEE(Main)Aug.2021]

फलन $f(x) = \log_{\sqrt{5}}\left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right)\right)$ का परिसर है :

[JEE(Main)Aug.2021]

(A) $[0, 2]$ (B) $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$ (C) $[-2, 2]$ (D) $\left(0, \sqrt{5}\right)$

Ans. A

Sol. $f(x) = \log_{\sqrt{5}}\left(3 + 2\cos\frac{\pi}{4}\cos x - 2\sin\frac{3\pi}{4}.\sin x\right)$

$= \log_{\sqrt{5}}\left(3 + \sqrt{2}\cos x - \sqrt{2}\sin x\right)$

$= \log_{\sqrt{5}}\left(3 + \sqrt{2}\left(\cos x - \sin x\right)\right)$

$-\sqrt{2} \le \cos x - \sin x \le \sqrt{2}$

$\Rightarrow -2 \le \sqrt{2}\left(\cos x - \sin x\right) \le \sqrt{2}$

$\Rightarrow 1 \le 3 + \sqrt{2}\left(\cos x - \sin x\right) \le 5$

$\log_{\sqrt{5}}(1) \le \log_{\sqrt{5}}(3 + \sqrt{2}\ (\cos x - \sin x)) \le \log_{\sqrt{5}} 5$.

$0 \le f(x) \le 2$

10. Let $f(x)$ be a polynomial of degree 3 such that $f(k) = -\frac{2}{k}$ for $k = 2, 3, 4, 5$. Then the value of $52 - 10f(10)$ is equal to _____ .

[JEE(Main)Aug.2021]

माना 3 घात का एक बहुपद $f(x)$ इस प्रकार है कि $k = 2, 3, 4, 5$ के लिए $f(k) = -\frac{2}{k}$ है। तब $52 - 10f(10)$ का मान के बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 26

Sol. $f(k) = -\frac{2}{k}$ for $k = 2, 3, 4, 5$

$\Rightarrow kf(k) + 2 = 0$

Consider the polynomial

$h(x) = xf(x) + 2$

$\because f(x)$ be polynomial of degree 3

$\because h(x)$ is polynomial of degree 4 r.t.

$h(2) = h(3) = h(4) = h(5) = 0$

$\because xf(x) + 2 = a(x-2)(x-3)(x-4)(x-5)$ .....(1)

Put $x = 10$

$\Rightarrow 10f(10) + 2 = 9 \times 8 \times 7 \times 6 \times 5$

$10f(10) + 2 = 1680a$

$52 - 10f(10) = 52 - 1680a + 2$

$= 54 - 1680a$ .....(2)

Put $x = 0$ in (1)

$2 = 120a$

$\because a = \frac{1}{60}$

$52 - 10f(10) = 54 - 28 \times \frac{1}{60}$

$= 54 - 28$

$= 26$

11. Let $f : N \to R$ be a function such that $f(x + y) = 2f(x)\, f(y)$ for natural numbers x and y. If $f(1) = 2$, then the value of α for which $\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}\left(2^{20} - 1\right)$ holds, is : **[JEE(Main)June2022]**

माना एक फलन $f : N \to R$ इस प्रकार है कि धन पूर्णांकों x तथा y के लिए $f(x + y) = 2f(x)\, f(y)$ है। यदि $f(1) = 2$ है, तो α का वह मान, जिसके लिए $\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}\left(2^{20} - 1\right)$ है, है : **[JEE(Main)June2022]**

(A) 2 (B) 3 (C) 4 (4) 6

Ans. C

Sol. $f: N \to R,\ f(x+y) = 2\,f(x)\,f(y)$ _____(1)

$f(1) = 2,$

$$\sum_{k=1}^{10} f(\alpha + k) = 2f(\alpha)\sum_{k=1}^{10} f(k)$$

$= 2f(\alpha)(f(1) + f(2) + ... + f(10))$ _____(2)

From (1)

$f(2) = 2\,f^2(1) = 2^3$

$f(3) = 2\,f(2)\,f(1) = 2^5$

$\vdots \qquad \vdots$

$f(10) = 2^9\,f^{10}(1) = 2^{19}$

$f(\alpha) = 2^{2\alpha - 1}\ ;\ \alpha \in N$

from (2)

$$\sum_{k=1}^{10} f(\alpha + k) = 2\left(2^{2\alpha-1}\right)\left(2 + 2^3 + 2^5 + ... + 2^{19}\right)$$

$$\frac{512}{3}\left(2^{20} - 1\right) = 2^{2\alpha}\left(2\frac{\left(2^{20} - 1\right)}{3}\right)$$

Hence $\alpha = 4$

12. Let $f(x) = \frac{x-1}{x+1}, x \in R - \{0, -1, 1\}$. If $f^{n+1}(x) = f\left(f^n(x)\right)$ for all $n \in N$, then $f^6(6) + f^7(7)$ is equal to :

[JEE(Main)June2022]

माना $f(x) = \frac{x-1}{x+1}, x \in R - \{0, -1, 1\}$ है। यदि $f^{n+1}(x) = f\left(f^n(x)\right)$, सभी $n \in N$ के लिए है, तो $f^6(6) + f^7(7)$ बराबर है :

[JEE(Main)June2022]

(A) $\frac{7}{6}$ (B) $-\frac{3}{2}$ (C) $\frac{7}{12}$ (4) $-\frac{11}{12}$

Ans. B

Sol. $f(x) = \frac{x-1}{x+1}$

$$\Rightarrow f^2(x) = f\left(f(x)\right) = \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1} = -\frac{1}{x}$$

$$f^3(x) = f\left(f^2(x)\right) = f\left(-\frac{1}{x}\right) = \frac{x+1}{1-x}$$

$$\Rightarrow f^4(x) = f\left(\frac{x+1}{1-x}\right) = -\frac{1}{x}$$

$$\Rightarrow f^6(x) = -\frac{1}{x} \Rightarrow f^6(6) = -\frac{1}{6}$$

$$f^7(x) = f\left(-\frac{1}{x}\right) = \frac{x+1}{1-x}$$

$$\Rightarrow f^7(7) = \frac{8}{-6} = -\frac{4}{3}$$

$$\therefore -\frac{1}{6} + -\frac{4}{3} = -\frac{3}{2}$$

13. Let $f: R \to R$ be defined as $f(x) = x - 1$ and $g: R - \{1, -1\} \to R$ be defined as $g(x) = \frac{x^2}{x^2 - 1}$. Then the function fog is : **[JEE(Main)June2022]**

(A) one-one but not onto (B) onto but not one-one

(C) both one-one and onto (4) neither one-one nor onto

माना $f: R \to R$ तथा $g: R - \{1, -1\} \to R$, $f(x) = x - 1$ तथा $g(x) = \frac{x^2}{x^2 - 1}$ द्वारा परिभाषित हैं। तो फलन fog :

[JEE(Main)June2022]

(A) एकैकी है परन्तु आच्छादक नहीं है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी तथा आच्छादक दोनों है (4) न तो एकैकी है न ही आच्छादक है

Ans. D

Sol. $f(x) = x - 1$ $g: R - \{1, -1\} \to R$

$f: R \to R$ $g(x) = \frac{x^2}{x^2 - 1}$

$$f(g(x)) = \frac{x^2 - 1}{x^2 - 1} = \frac{x^2 - x^2 + 1}{x^2 - 1}$$

$f(g(x)) = \frac{1}{x^2 - 1}$ $fog: R - \{1, -1\} \to R$

Neither one–one nor onto

14. Let a function $f: N \to N$ be defined dy

$$f(n)=\begin{cases}2n, & n=2,4,6,8,.....\\ n-1, & n=3,7,11,15,.....\\ \frac{n+1}{2}, & n=1,5,9,13,.....\end{cases}$$

Then, f is : **[JEE(Main)June2022]**

(A) one-one but not onto (B) onto but not one-one

(C) neither one-one nor onto (4) one-one and onto

माना एक फलन $f: N \to N$

$$f(n)=\begin{cases}2n, & n=2,4,6,8,.....\\ n-1, & n=3,7,11,15,.....\\ \frac{n+1}{2}, & n=1,5,9,13,.....\end{cases}$$

द्वारा परिभाषित है। तो f : **[JEE(Main)June2022]**

(A) एकैकी है परन्तु आच्छादक नहीं है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) न तो एकैकी है न ही आच्छादक है (4) एकैकी तथा आच्छादक है

Ans. D

Sol. f(x) = 1 for n = 1
2 for n = 3
3 for n = 5
4 for n = 2
So onto and one-one

15. The number of one-one functions $f: \{a, b, c, d\} \to \{0, 1, 2, ...,10\}$ such that $2f(a) - f(b) + 3f(c) + f(d) = 0$ is _____. **[JEE(Main)June2022]**

एकैकी फलनों $f: \{a, b, c, d\} \to \{0, 1, 2, ...,10\}$, जिनके लिए $2f(a) - f(b) + 3f(c) + f(d) = 0$ है, की संख्या है _____ **[JEE(Main)June2022]**

Ans. 31

Sol. $\because 3f(c) + 2f(a) + f(d) = f(b)$

Value of f(c)	Value of f(a)	Number of functions
0	1	7
	2	5
	3	3
	4	2
1	0	6
	2	2
	3	1
2	0	3
	1	1
3	0	1
	Total Number of functions =	**31**

16. Let $f: R \to R$ be a function defined by $f(x)=\left(2\left(1-\frac{x^{25}}{2}\right)\left(2+x^{25}\right)\right)^{\frac{1}{50}}$. If the function $g(x)=f\left(f\left(f(x)\right)\right)+f\left(f(x)\right)$, then the greatest integer less than or equal or equal to g(1) is _____.

[JEE(Main)June2022]

माना फलन $f: R \to R$, $f(x)=\left(2\left(1-\frac{x^{25}}{2}\right)\left(2+x^{25}\right)\right)^{\frac{1}{50}}$ द्वारा परिभाषित है। यदि फलन $g(x)=f\left(f\left(f(x)\right)\right)+f\left(f(x)\right)$ है, तो g(1) के बराबर या उससे कम महत्तम पूर्णांक _____ है। **[JEE(Main)June2022]**

Ans. B

Sol. $f(x)=\left[\left(2-x^{25}\right)\left(2+x^{25}\right)\right]^{\frac{1}{50}}$

$=\left(4-x^{50}\right)^{1/50}$

$f(f(x))=\left(4-\left(\left(4-x^{50}\right)^{1/50}\right)^{50}\right)^{1/50}=x$

$g(x)=f(f(f(x)))+f(f(x))$

$=f(x)+x$

$g(1)=f(1)+1=3^{1/50}+1$

$[g(1)]=\left[3^{1/50}+1\right]=2$

17. Let $f: R \to R$ be a function defined by $f(x)=\frac{2e^{2x}}{e^{2x}+e}$. Then $f\left(\frac{1}{100}\right)+f\left(\frac{2}{100}\right)+f\left(\frac{3}{100}\right)+....+f\left(\frac{99}{100}\right)$ is equal to _____.

[JEE(Main)June2022]

माना फलन $f: R \to R,\ f(x) = \frac{2e^{2x}}{e^{2x}+e}$ द्वारा परिभाषित है। तो $f\left(\frac{1}{100}\right)+f\left(\frac{2}{100}\right)+f\left(\frac{3}{100}\right)+....+f\left(\frac{99}{100}\right)$ बराबर है _____

[JEE(Main)June2022]

Ans. 99

Sol. $f(x) = \frac{2e^{2x}}{e^{2x}+e}$

$f(1-x) = \frac{2e^{2-2x}}{e^{2-2x}+e}$

$f(1-x) = \frac{2e^{2-2x}}{e^{2-2x}+e} = \frac{(2e)}{(e+e^{2x})}$

$f(x)+f(1-x) = \frac{2e^{2x}}{e^{2x}+e} + \frac{2e}{e^{2x}+e} = 2$

$f\left(\frac{1}{100}\right)+f\left(\frac{2}{100}\right)+f\left(\frac{3}{100}\right).....f\left(\frac{99}{100}\right)$

$f\left(\frac{1}{100}\right)+f\left(\frac{99}{100}\right) = 2$

$f\left(\frac{2}{100}\right)+f\left(\frac{98}{100}\right) = 2$

$\vdots$

$f\left(\frac{49}{100}\right)+f\left(\frac{51}{100}\right) = 2$

$f\left(\frac{50}{100}\right) = f\left(\frac{1}{2}\right)$

$= 49 \times 2 + f\left(\frac{1}{2}\right) = 1$

$= 98 + 1 = 99$

18. let S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Define f: S → S as $f(n) = \begin{cases} 2n, \text{if } n = 1,2,3,4,5 \\ 2n-11, \text{if } n = 6,7,8,9,10 \end{cases}$. Let g : S → S be function such that $fog(n) = \begin{cases} n+1, \text{if n is odd} \\ n-1, \text{if n is even} \end{cases}$. Then g(10) ((g(1) + g(2) +g(3) + g(4) + g(5)) is equal to _____.

[JEE(Main)June2022]

माना S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} है। f: S $\to$ S को $f(n)=\begin{cases}2n, \text{if } n=1,2,3,4,5\\ 2n-11, \text{if } n=6,7,8,9,10\end{cases}$ द्वारा परिभाषित कीजिए।

माना एक फलन g : S $\to$ S के लिए $fog(n)=\begin{cases}n+1, \text{य दि n विषम है}\\ n-1, \text{य दि n सम है}\end{cases}$ तो g(10) ((g(1) + g(2) +g(3) + g(4) + g(5)) बराबर है _____. **[JEE(Main)June2022]**

Ans. 190

Sol. $g(n)=\begin{cases}f^{-1}(n+1) & , \quad n\in \text{odd}\\ f^{-1}(n-1) & , \quad n\in \text{even}\end{cases}$

$f^{-1}(n)=\begin{cases}\dfrac{n}{2} & , \quad n=2,4,6,8,10\\ \dfrac{n+11}{2} & , \quad n=1,3,5,7,9\end{cases}$

$g(n)=\begin{cases}\dfrac{n+1}{2} & , \quad n\in \text{odd}\\ \dfrac{n+10}{2} & , \quad n\in \text{even}\end{cases}$

$g(10)=10, \quad g(1)=1, \quad g(2)=6, \quad g(3)=2, \quad g(4)=7 \quad g(5)=3$

$\Rightarrow$ 190 Ans.

19. Let S = {1, 2, 3, 4}. Then the number of elements in the set {f : S × S $\to$ S : f is onto and $f(a,b)=f(b,a)\ge a\forall(a,b)\in S\times S$ } is _____. **[JEE(Main)June2022]**

माना S = {1, 2, 3, 4} है। तो समुच्चय {f : S × S $\to$ S : f आच्छादक है तथा $f(a,b)=f(b,a)\ge a\forall(a,b)\in S\times S$ }में अवयवों की संख्या है _____ **[JEE(Main)June2022]**

Ans. 37

Sol. There are 16 ordered pass in 5 × 5. We write all these ordered pairs in 4 sets as follows :

A = {(1, 1)}

B = {(1, 4), (2, 4), (3, 4), (4, 4), (4, 3), (4, 2), (4, 1)}

C = {(1, 3), (2, 3), (3, 3), (3, 2), (3, 1)}

D = {(1, 2), (2, 2), (2, 1)}

All elements of set B have image 4 and element of A has only image 1.

All elements of set C have image 3 or 4 and all elements of set D have image 2 or 3 or 4.

Case - I : When no element of set C has image 3 number of onto functions = 2 (when elements of set D have images 2 or 3)

Case - II : When at least one element of set C has image 3. Number of onto functions = $(2^3 - 1)(1 + 2 + 2)$

$= 35$

Total number of functions = 37.

20. Let c, k ∈ R. If $f(x) = (c + 1)x^2 + (1 - c^2)x + 2k$ and $f(x + y) = f(x) + f(y) - xy$, for all x, y ∈ R, then the value of $|2(f(1) + f(2) + f(3) + ... + f(20))|$ is equal to _____. **[JEE(Main)June2022]**

माना c, k ∈ R हैं। यदि सभी x, y ∈ R के लिए $f(x) = (c + 1)x^2 + (1 - c^2)x + 2k$ तथा $f(x + y) = f(x) + f(y) - xy$, हैं, तो $|2(f(1) + f(2) + f(3) + ... + f(20))|$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 3395

Sol. f(x) is a polynomial

s.t. $f(x) = (c+1)x^2 + (1-c^2)x + 2k$ .....(i)

and $f(x + y) = f(x) + f(y) - xy \quad x, y \in R$ .....(ii)

put x = y = 0 in (2)

$f(0) = 0 \Rightarrow$ from (1) $k = 0$

put x = y

$2f(x) - f(2x) = x^2$

$2(c+1)x^2 + 2(1+c^2)x - (c+1)4x^2 - (1-c^2)2x = x^2$

Compair coefficient of $x^2 \Rightarrow -2(1 + c) = 1$

$$C = \frac{-3}{2}$$

from (1) $2f(x) = -x^2 - \frac{5}{2}x$

$$\left|2\left[f(1)+f(2)+....+f(20)\right]\right| = \left\{\frac{20(21)(41)}{6} + \frac{5}{2}\frac{20(21)}{2}\right\} = 3395$$

21. Let f(x) and g(x) be two polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and $g(f(x)) = 4x^2 + 6x + 1$, then the value of f(2) + g(2) is _____. **[JEE(Main)June2022]**

माना दो वास्तविक बहुपद f(x) तथा g(x) क्रमशः घात 2 तथा 1 के हैं। यदि $f(g(x)) = 8x^2 - 2x$ तथा $g(f(x)) = 4x^2 + 6x + 1$ हैं, तो f(2) + g(2) का मान है _____ **[JEE(Main)June2022]**

Ans. 18

Sol. $f(x) = ax^2 + bx + c$

$g(x) = dx + e$

$fog(x) = a(g(x))^2 + bg(x) + c$

$= a[d^2x^2 + e^2 + 2dex] + b[dx + e] + c$

$= ad^2x^2 + x(2ade + bd) + e^2 + be + c$

gof(x)

$= df(x) + e$

$= adx^2 + bdx + cd + e$

on comparing

$ad = 4, bd = 6, cd + e = 1$

$ad^2 = 8, 2ade + bd = -2$

$e^2 + be + c = 0$

$ae = -2$

On solving

$a = 2, b = 3, c = 1, d = 2, e = -1$

$f(x) = 2x^2 + 3x + 1$

$g(x) = 2x - 1$

22. The number of functions, $f: \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$ such that $f(1) + f(2) = f(3)$, is equal to :

[JEE(Main)July2022]

फलनों $f: \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$, जिनके लिए $f(1) + f(2) = f(3)$ है, की कुल संख्या है :

[JEE(Main)July2022]

(A) 60 (B) 90 (C) 108 (4) 126

Ans. B

Sol. $A = \{1, 2, 3, 4\}$

$B = \{1, 2, 3, 4, 5, 6\}$

Here f(3) can be 2, 3, 4, 5, 6

$f(3) = 2, (f(1), f(2)) \to (1,1) \to 6$ cases

$f(3) = 3, (f(1), f(2)) \to (1,2), (2,1)$

$\to 2 \times 6 = 12$ cases

$f(3) = 4, (f(1), f(2)) \to (1,3), (3,1), (2,2)$

$\to 3 \times 6 = 18$ cases

$f(3) = 5, (f(1), f(2)) \to (1,4), (4,1), (2,3), (3,2)$

$\to 4 \times 6 = 24$ cases

$f(3) = 6, (f(1), f(2)) \to (1,5),(5,1),(2,4),(4,2),(3,3)$

$\to 5 \times 6 = 30$ cases

Total number of cases = 6+ 12 + 18 + 24 + 30 = 90

23. The number of bijective functions $f: \{1, 3, 5, 7, ..., 99\} \to \{2, 4, 6, 8, ..., 100\}$, such that $f(3) \geq f(9) \geq f(15) \geq f(21) \geq ..... \geq f(99)$, is : **[JEE(Main)July2022]**

एकैकी आच्छादी फलनों $f: \{1, 3, 5, 7, ..., 99\} \to \{2, 4, 6, 8, ..., 100\}$ जिनके लिए

$f(3) \geq f(9) \geq f(15) \geq f(21) \geq ..... \geq f(99)$, हैं, की संख्या है : **[JEE(Main)July2022]**

(A) ${}^{50}P_{17}$ (B) ${}^{50}P_{33}$ (C) $33! \times 17!$ (4) $\frac{50!}{2}$

Ans. B

Sol. Let number of terms in sequence

$\Rightarrow 3,9,15, ...,99$ be n

$\Rightarrow 99 = 3 + (n - 1)\, 6 \Rightarrow n = 17$

number of bijective functions = ${}^{50}C_{17}.33!$

$= {}^{50}C_{33}.33! = {}^{50}P_{33}$

24. Let $f, g : N - \{1\} \to N$ be functions defined by $f(a) = \alpha$, where α is the maximum of the powers of those primes p such that p^α divides a, and $g(a) = a + 1$, for all $a \in N - \{1\}$. Then, the function $f + g$ is :

[JEE(Main)July2022]

(A) one-one but not onto (B) onto one-one and onto

(C) both one-one and onto (4) neither one-one nor onto

माना $f, g : N - \{1\} \to N$, $f(a) = \alpha$, जहाँ उन अभाज्य संख्याओं p, जिनके लिए p^α , a को विभाजित करता है, की घातों में α अधिकतम है तथा $g(a) = a + 1$ सभी $a \in N - \{1\}$ के लिए द्वारा परिभाषित हैं। तो फलन $f + g$:

[JEE(Main)July2022]

(A) एकैकी है परन्तु आच्छादक नहीं है (B) आच्छादक है परन्तु एकैकी नहीं है

(C) एकैकी तथा आच्छादक दोनों है (4) न तो एकैकी है न ही आच्छादक

Ans. D

Sol. $f : N - \{1\} \to N$ $f(a) = \alpha$

Where α is max of powers of prime P such that p^α divides a. Also $g(a) = a + 1$

$\therefore$ $f(2) = 1$ $g(2) = 3$

$f(3) = 1$ $g(3) = 4$

$f(4) = 2$ $g(4) = 5$

$f(5) = 1$ $g(5) = 6$

$\Rightarrow$ $f(2) + g(2) = 4$

$(f(3) + g(3)) = 5$

$f(4) + g(4) = 7$

$f(5) + g(5) = 7$

$\therefore$ Many one $f(x) + g(x)$ does not cotain 1

$\Rightarrow$ into function

$\therefore$ Ans. (D) [neither one-one nor onto]

25. The domain of the function $f(x) = \sin^{-1}\left[2x^2 - 3\right] + \log_2\left(\log_{\frac{1}{2}}\left(x^2 - 5x + 5\right)\right)$, where [t] is the greatest integer function, is :

[JEE(Main)July2022]

फलन $f(x) = \sin^{-1}\left[2x^2 - 3\right] + \log_2\left(\log_{\frac{1}{2}}\left(x^2 - 5x + 5\right)\right)$, जहाँ [t] महत्तम पूर्णांक फलन है, का प्रांत है :

[JEE(Main)July2022]

(A) $\left(-\sqrt{\frac{5}{2}}, \frac{5-\sqrt{5}}{2}\right)$ (B) $\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$

(C) $\left(1, \frac{5-\sqrt{5}}{2}\right)$ (4) $\left[1, \frac{5+\sqrt{3}}{2}\right)$

Ans. C

Sol. $f(x)=\sin^{-1}[2x^2-3]+\log_2\left(\log_{\frac{1}{2}}\left(x^2-5x+5\right)\right)$

$P_1 : -1 \le [2x^2-2] < 1$

$\Rightarrow -1 \le 2x^2-3 < 2$

$\Rightarrow 2 < 2x^2 < 5$

$\Rightarrow 1 < x^2 < \frac{5}{2}$

$\Rightarrow P_1 : x \in \left(-\sqrt{\frac{5}{2}}, -1\right) \cup \left(1, \sqrt{\frac{5}{2}}\right)$

$P_2 : x^2 - 5x + 5 > 0$

$\Rightarrow \left(x-\left(\frac{5-\sqrt{5}}{2}\right)\right)\left(x-\left(\frac{5-\sqrt{5}}{2}\right)\right) > 0$

$P_3 : \log_{\frac{1}{2}}\left(x^2-5x+5\right) > 0$

$\Rightarrow x^2-5x-5 < 1$

$\Rightarrow x^2-5x+4 < 0$

$\Rightarrow P_3 : x \in (1,4)$

So, $P_1 \cap P_2 \cap P_3 = \left(1, \frac{5-\sqrt{5}}{2}\right)$

26. Let $S=\left\{x \in [-6,3]-\{-2,2\} : \frac{|x+3|-1}{|x|-2} \ge 0\right\}$ and $T = \{x \in Z : x^2 - 7|x| + 9 \le 0\}$. Then the number of elements in $S \cap T$ is : **[JEE(Main)July2022]**

माना $S=\left\{x \in [-6,3]-\{-2,2\} : \frac{|x+3|-1}{|x|-2} \ge 0\right\}$ तथा $T = \{x \in Z : x^2 - 7|x| + 9 \le 0\}$ हैं। तो $S \cap T$ में अवयवों की संख्या है : **[JEE(Main)July2022]**

(A) 7 (B) 5 (C) 4 (4) 3

Ans. D

Sol. $S \cap T = \{-5, -4, 3\}$

27. Let $f(x) = ax^2 + bx + c$ be such that $f(1) = 3$, $f(-2) = \lambda$ and $f(3) = 4$. If $f(0) + f(1) + f(-2) + f(3) = 14$, then λ is equal to : **[JEE(Main)July2022]**

माना $f(x) = ax^2 + bx + c$ है, जिसके लिए $f(1) = 3$, $f(-2) = \lambda$ तथा $f(3) = 4$ हैं। यदि $f(0) + f(1) + f(-2) + f(3) = 14$ है, तो λ बराबर है : **[JEE(Main)July2022]**

(A) -4 (B) $\frac{13}{2}$ (C) $\frac{23}{2}$ (4) 4

Ans. D

Sol. $f(0) + 3 + \lambda + 4 = 14$

$\therefore f(0) = 7 - \lambda = c$

$f(1) = a + b + c = 3$ .....(i)

$f(3) = 9a + 3b + c = 4$ .....(ii)

$f(-2) = 4a - 2b + c = l$ .....(iii)

(ii) – (iii)

$a + b = \frac{4-\lambda}{4}$ put in equation (i)

$\frac{4-\lambda}{5} + 7 - \lambda = 3$

$6\lambda = 24; \quad \lambda = 4$

28. The number of functions f, from the set $A = \{x \in N : x^2 - 10x + 9 \le 0\}$ to the set $B = \{n^2 : n \in N\}$ such that $f(x) \le (x-3)^2 + 1$, for every $x \in A$, is _____. **[JEE(Main)July2022]**

समुच्चय $A = \{x \in N : x^2 - 10x + 9 \le 0\}$ से समुच्चय $B = \{n^2 : n \in N\}$ ऐसे फलनों, जिनके लिए $f(x) \le (x-3)^2 + 1$, $x \in A$ है, की संख्या है _____ **[JEE(Main)July2022]**

Ans. 1440

Sol. $(x^2 - 10x + 9) \le T\ 0 \Rightarrow (x-1)(x-9) \le 0$

$\Rightarrow x \in [1, 9] \Rightarrow A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$f(x) \le (x-3)^2 + 1$

$x = 1 : f(1) \le 5 \Rightarrow 1^2, 2^2$

$x = 2 : f(2) \le 2 \Rightarrow 1^2$

$x = 3 : f(3) \le 1 \Rightarrow 1^2$

$x = 4 : f(4) \le 2 \Rightarrow 1^2$

$x = 5 : f(5) \le 5 \Rightarrow 1^2, 2^2$

$x = 6 : f(6) \le 10 \Rightarrow 1^2, 2^2, 3^2$

$x = 7 : f(7) \le 17 \Rightarrow 1^2, 2^2, 3^2, 4^2$

$x = 8 : f(8) \le 26 \Rightarrow 1^2, 2^2, 3^2, 4^2, 5^2$

$x = 9 : f(9) \le 37 \Rightarrow 1^2, 2^2, 3^2, 4^2, 5^2, 6^2$

Total number of such function

$= 2(6!) = 2(720) = 1440$